The complete excess-two inverse decoder layer

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For a positive odd integer x, let

$$U_+(x) := \frac{3x+1}{2^{v_2(3x+1)}}. \tag{1}$$

An exponent word $E = (e_1, \ldots, e_h)$ is read from its positive odd start toward its terminal state. Its mass and excess are

$$\mathrm{B}(E) := e_1 + \cdots + e_h, \qquad t(E) := \mathrm{B}(E) - h. \tag{2}$$

This note classifies every exact inverse word with $t(E) = 2$ at a fixed positive odd terminal state.

The elementary reverse rule is useful. If x is the current odd terminal state, an exponent-a predecessor is

$$p = \frac{2^a x - 1}{3}. \tag{3}$$

It exists precisely when $3 \nmid x$ and

$$a \equiv \begin{cases} 1 \pmod 2, & x \equiv 2 \pmod 3, \\ 0 \pmod 2, & x \equiv 1 \pmod 3. \end{cases} \tag{4}$$

In shifted coordinates $X = x + 1$, reversing exponents one, two, and three uses respectively

$$X \longmapsto \frac{2X}{3}, \qquad X \longmapsto \frac{4X-2}{3}, \qquad X \longmapsto \frac{8X-6}{3}. \tag{5}$$

Theorem 1 (complete excess-two classification)**.** *Let r be a positive odd integer and put*

$$b := v_3(r+1), \qquad s := \frac{r+1}{3^b}. \tag{6}$$

There are exactly two families of exact predecessor words E with $t(E) = 2$.

Family I: one exponent three. *For every k with $0 \le k < b$, define*

$$M_k := 2^{k+2}3^{b-k-1}s - 1, \qquad c_k := v_3(M_k). \tag{7}$$

For every $0 \le \ell \le c_k$, there is one exact word

$$\boxed{E = 1^\ell 3\, 1^k, \qquad |E| = k + 1 + \ell, \qquad z + 1 = 2\left(\frac{2}{3}\right)^\ell M_k.} \tag{8}$$

These are all the excess-two words containing an exponent three.

Family II: two exponents two. *Define*

$$N_1 := 2^{b+1}s - 1, \qquad b_1 := v_3(N_1). \tag{9}$$

This family is empty if $b_1 = 0$. If $b_1 \ge 1$, put

$$s_1 := \frac{N_1}{3^{b_1}}, \qquad N_2 := 2^{b_1+1}s_1 - 1, \qquad b_2 := v_3(N_2). \tag{10}$$

For every $1 \le j \le b_2$, there is one exact word

$$\boxed{E = 1^{j-1}2\, 1^{b_1-1}2\, 1^b, \qquad |E| = b + b_1 + j, \qquad z + 1 = \left(\frac{2}{3}\right)^j N_2.} \tag{11}$$

These are all the excess-two words containing two exponents two. There are no other exact words of excess two.

Proof. Because all exponents are positive and their excesses sum to two, the word has exactly one of the two forms indicated: one exponent three and all other exponents one, or two exponents two and all other exponents one.

Read the word backward from r, and let k be the number of trailing exponents one. Rule (5) shows that precisely b consecutive exponent-one inverses are available at the outset.

First suppose that the next non-one exponent is three. Its parity in (4) is odd, so it must occur before the exponent-one run is exhausted; thus $0 \le k < b$. After reversing the k trailing ones, the shifted state is

$$X = 2^k 3^{b-k} s. \tag{12}$$

Reversing exponent three gives

$$\frac{8X - 6}{3} = 2\big(2^{k+2}3^{b-k-1}s - 1\big) = 2M_k. \tag{13}$$

All remaining exponents must equal one. Exactly $c_k = v_3(M_k)$ of them are available, and ℓ such steps give shifted start

$$\left(\frac{2}{3}\right)^{\ell} 2M_k.$$

This proves (7)–(8), both existence and exhaustion of Family I.

Now suppose that the word contains two exponents two. Even parity in (4) forces the first such exponent, read backward, to occur only after all b initial exponent-one inverses. At that point the shifted state is $2^b s$, and reversing exponent two would give

$$\frac{4\,2^b s - 2}{3} = \frac{2N_1}{3}. \tag{14}$$

It is possible exactly when $b_1 \geq 1$. The shifted state in (14) has 3-adic valuation $b_1 - 1$. Since the only remaining unit of excess must form another exponent two, parity forces the intervening run to have exactly $b_1 - 1$ exponents one. After that run the shifted state is

$$2^{b_1} s_1. \tag{15}$$

The second exponent-two inverse gives $2N_2/3$ and is possible exactly when $b_2 \geq 1$. Thereafter all exponents equal one; reversing $j-1$ of them is possible precisely for $1 \leq j \leq b_2$, and gives shifted start

$$\left(\frac{2}{3}\right)^{j-1} \frac{2N_2}{3} = \left(\frac{2}{3}\right)^{j} N_2. \tag{16}$$

The word and length in (11) follow. The parity alternatives above are forced at every first non-one position, so the two families exhaust all excess-two words. □

Corollary 2 (finite excess-two horizon)**.** *With the notation of Theorem 1, set*

$$H_I := \max_{0 \leq k < b} (k + 1 + c_k), \tag{17}$$

where the maximum of the empty set is zero, and set

$$H_{II} := \begin{cases} b + b_1 + b_2, & b_1 \geq 1 \text{ and } b_2 \geq 1, \\ 0, & b_1 b_2 = 0. \end{cases} \tag{18}$$

Then no exact h-step predecessor word of r has excess two when

$$\boxed{h > H_2(r) := \max(H_I, H_{II}).} \tag{19}$$

Combining this with the excess-zero and excess-one decoder theorem, put

$$H_{\le 2}(r) := \max\bigl(b,\ b + b_1,\ H_2(r)\bigr). \tag{20}$$

Every exact h-step predecessor word with $h > H_{\le 2}(r)$ has excess at least three.

Proof. The largest length in Family I is (17), and the largest possible length in Family II is (18). This proves (19). Excess zero ends at depth b, and excess one ends at depth $b + b_1$. Taking the maximum proves (20). □

Corollary 3 (fixed-seed inequalities)**.** *Let n be the globally least positive odd integer with an unbounded accelerated orbit, and suppose the orbit of r is unbounded. Then every Family I branch satisfies*

$$\boxed{M_k \ge \frac{n+1}{2}\left(\frac{3}{2}\right)^{\ell} \qquad (0 \le \ell \le c_k),} \tag{21}$$

and every Family II branch satisfies

$$\boxed{N_2 \ge (n+1)\left(\frac{3}{2}\right)^{j} \qquad (1 \le j \le b_2).} \tag{22}$$

In particular, at the least same-depth suffix q_* of a record-atom block, all variable-depth prefixes of excess at most two are now given by finite explicit valuation ladders. Beyond $H_{\le 2}(q_*)$, the minimum suffix channel has prefix excess at least three.

Proof. Every positive predecessor of r also has an unbounded forward orbit, so its start z is at least n. Substitution of the two exact shifted starts in (8) and (11) gives (21) and (22). The final assertion is Corollary 2 applied to $r = q_*$. □